MATH 123

Homework 0 Demo

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Problem 1 (Analysis). Let $(X, \mathcal{F}, \mu)$ be a measure space and $\{f_n\}_{n=1}^{\infty}$ a sequence of measurable functions $f_n : X \to \mathbb{R}$.

(a) Suppose $\sum_{n=1}^{\infty} \int_{\mathbb{R}} |f_n| \, \mathrm{d}\mu < \infty$. Prove that

$$\int_{\mathbb{R}} \sum_{n=1}^{\infty} f_n \, \mathrm{d}\mu \; = \; \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n \, \mathrm{d}\mu,$$

i.e. that integration and summation may be interchanged.

(b) Let $a_n \geq 0$ and suppose $\sum_{n=1}^{\infty} a_n < \infty$. Using the inequality $\log(1+t) \leq t$ for $t \geq 0$, show that the infinite product $\prod_{n=1}^{\infty}(1+a_n)$ converges, and that

$$\prod_{n=1}^{\infty}(1+a_n) \; \leq \; \exp(\sum_{n=1}^{\infty} a_n) \; < \; \infty.$$

Solution. (a) Define $g := \sum_{n=1}^{\infty} |f_n|$. By the Monotone Convergence Theorem applied to the partial sums $g_N = \sum_{n=1}^{N} |f_n| \nearrow g$,

$$\int_X g \, \mathrm{d}\mu \; = \; \int_X \sum_{n=1}^{\infty} |f_n| \, \mathrm{d}\mu \; = \; \lim_{n \to \infty} \sum_{k=1}^{n} \int_X |f_k| \, \mathrm{d}\mu \; = \; \sum_{n=1}^{\infty} \int_X |f_n| \, \mathrm{d}\mu \; < \; \infty.$$

Hence $g \in L^1(\mu)$, so $|\sum_{n=1}^{\infty} f_n| \leq g$ provides an integrable dominating function. Applying the Dominated Convergence Theorem to the partial sums $S_N := \sum_{n=1}^{N} f_n \to \sum_{n=1}^{\infty} f_n$ pointwise,

$$\int_X \sum_{n=1}^{\infty} f_n \, \mathrm{d}\mu \; = \; \lim_{n \to \infty} \int_X \sum_{k=1}^{n} f_k \, \mathrm{d}\mu \; = \; \lim_{n \to \infty} \sum_{k=1}^{n} \int_X f_k \, \mathrm{d}\mu \; = \; \sum_{n=1}^{\infty} \int_X f_n \, \mathrm{d}\mu. \quad \square$$

(b) For each finite partial product, taking logarithms and applying $\log(1+t) \leq t$,

$$\log \prod_{n=1}^{N}(1+a_n) \; = \; \sum_{n=1}^{N} \log(1+a_n) \; \leq \; \sum_{n=1}^{N} a_n \; \leq \; \sum_{n=1}^{\infty} a_n \; < \; \infty.$$

Since $a_n \geq 0$, the partial products are monotone increasing and bounded above by $\exp(\sum_{n=1}^{\infty} a_n)$, hence converge. Taking $N \to \infty$,

$$\prod_{n=1}^{\infty}(1+a_n) \; \leq \; \exp(\sum_{n=1}^{\infty} a_n) \; < \; \infty.$$

$\square$

Problem 2 (Abstract Algebra). Recall that a sequence of R-modules

$$0 \longrightarrow A \xrightarrow{\iota} B \xrightarrow{\pi} C \longrightarrow 0$$

is *short exact* if $\iota : A \hookrightarrow B$ is injective, $\pi : B \twoheadrightarrow C$ is surjective, and $\mathrm{im}(\iota) = \ker(\pi)$.

(a) Prove that the sequence splits — i.e. $B \cong A \oplus C$ compatibly with ι and π — if and only if:

(i) there exists an R-module homomorphism $\sigma : C \hookrightarrow B$ such that $\pi \circ \sigma = \mathrm{id}_C$ (a *section* of π), and equivalently,

(ii) there exists an R-module homomorphism $\rho : B \twoheadrightarrow A$ such that $\rho \circ \iota = \mathrm{id}_A$ (a *retraction* of ι).

Construct the splitting isomorphism $\phi : B \xrightarrow{\sim} A \oplus C$ explicitly in each case.

(b) State the Structure Theorem for finitely generated modules over a PID R, and identify the two canonical forms it produces.

Solution. (a) We prove both equivalences; by symmetry it suffices to show each direction for (i), then observe (ii) is entirely dual.

Section σ gives splitting. Suppose $\sigma : C \hookrightarrow B$ satisfies $\pi \circ \sigma = \mathrm{id}_C$. Define

$$\phi : B \longrightarrow A \oplus C, \qquad \phi(b) \;:=\; \big(b - \sigma(\pi(b)),\ \pi(b)\big),$$

where we regard $A \hookrightarrow B$ via ι (exactness makes this unambiguous). ϕ is R-linear since π and σ are.

Injective: if $\phi(b) = 0$ then $\pi(b) = 0$ and $b = \sigma(0) = 0$.
Surjective: given $(a, c) \in A \oplus C$, set $b = \iota(a) + \sigma(c)$; one checks $\phi(b) = (a, c)$.

Hence $\phi : B \xrightarrow{\sim} A \oplus C$, and by construction $\phi \circ \iota = \iota_A$ and $\pi_C \circ \phi = \pi$, so the splitting is compatible. □

Splitting gives section. If $\phi : B \xrightarrow{\sim} A \oplus C$ compatibly, set $\sigma := \phi^{-1} \circ \iota_C : C \hookrightarrow B$ where $\iota_C : C \hookrightarrow A \oplus C$ is the canonical inclusion. Then $\pi \circ \sigma = \pi_C \circ \phi \circ \phi^{-1} \circ \iota_C = \mathrm{id}_C$. □

Retraction ρ gives splitting. Define $\psi : A \oplus C \longrightarrow B$ by choosing any lift $\tilde{b} \in \pi^{-1}(c)$ and setting

$$\psi(a, c) \;:=\; \iota(a) + \tilde{b} - \iota(\rho(\tilde{b})).$$

The term $\tilde{b} - \iota(\rho(\tilde{b}))$ lies in $\ker \rho = \mathrm{im}\, \iota$ and depends only on c (not the choice of lift), so ψ is well-defined and R-linear. One verifies $b \mapsto (\rho(b), \pi(b))$ is its inverse, giving $B \xrightarrow{\sim} A \oplus C$. □

(b) Let M be a finitely generated module over a PID R. Then

$$M \;\cong\; R^r \oplus \bigoplus_{i=1}^{k} R/\left\langle d_i \right\rangle,$$

where $r \geq 0$ is the *free rank* and $d_1 \mid d_2 \mid \cdots \mid d_k$ are nonzero non-unit *invariant factors* in R (**invariant factor form**). Equivalently, each torsion summand decomposes into primary pieces $R/\left\langle p^e \right\rangle$ via CRT, giving the **primary decomposition form**. Both are unique up to associates.

□